



## Software Engineering - Technical Assignment

### Business focus:

*In order to comply to the customer protection act, any credit institution in Finland must assess customer's credit worthiness on a fair, common and transparent ground, to prevent households from taking on too much debt. The Finnish Tax Administration's Incomes Register Unit controls and provides the information on the Positive Credit Register.*

### High level requirement:

1. It must be possible to fetch the credit extract information from the Positive Credit Register on the consumer with a financing need.
2. It must be possible to view the detail credit extract information of the consumer for all the instances fetched from the Positive Credit Register.
3. It must be possible to monitor and report on the consumer financing requests for the consumers with 'voluntary credit ban' option activated.



# Software Engineering - Technical Assignment

## Assignment tasks:

Your task is to design and build a solution that fulfills the given high level requirement. Please follow the below technical implementation instructions and provide the relevant deliverables.

### Technical implementation:

- Implement a minimal, working proof-of-concept, reflecting the different key elements in at least 3 major components - reflect modern microservices patterns:
  - a front-end for user interaction;
  - a back-end for data fetching and storage;
  - a monitoring/reporting service for voluntary credit ban.
- You may choose your preferred tech stack (React/Java would be recommended) - corresponding libraries and frameworks are free of choice.
- The solution should be tested moderately and error handling should be illustrated.
- Demonstrate how the solution can be setup, executed and deployed.
- Write a concise README.md file that describes your solution, assumptions and trade-offs made during the development.

#### 1. A front-end for user interaction

- user can trigger a new fetch from the Positive Credit Register for a given consumer SSN.
- user can view the fetch history on a given consumer.
- user can view the detail information of each fetch instance from the Positive Credit Register.

#### 2. A back-end for data fetching and storage

- an API integration to a mocked REST API based on the [API documentation](#) provided by the Positive Credit Register. (see more technical documentation [here](#))
- a data persistence of your choice to store the relevant data.
- APIs to be consumed by the front-end.
- APIs to enable the monitoring/reporting service.

#### 3. A monitoring/reporting service for voluntary credit ban

- user should be notified via email ([pcr\\_monitoring@dansketest.dk](mailto:pcr_monitoring@dansketest.dk)) on a regular basis.
- the form of the report is free of choice - e.g.: plain text, file, web report etc.



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## Assignment tasks:

### Deliverables

Please submit a single link to a Git repository (e.g., GitHub, GitLab) containing:

1. Solution elements required.
2. [Optional] A 5-minute solution walk-through video where you walk through your code, diagram, and demo the working system.

### Evaluation Criteria

- System Design: Clean separation of concerns between the frontend, backend, and among the microservice functionalities.
- Code Quality: Readability, modularity, and adherence to language-specific best practices.
- Resilience: How the system handles failures (e.g., what happens if the notification service is down when a credit ban is detected?).
- Documentation: Clarity of the README and ease of getting the environment running.

### Note to the Candidate

We understand that your time is valuable. The goal is not to build a perfect, production-grade system but to demonstrate your thought process and technical capabilities along with your business understanding and problem-solving skills.

Focus on the core functionalities, instead of everything/edge cases to be time efficient, and please keep a track of the time spent and document the effort accordingly.

You are welcome to use any tool at hand to accelerate development.

We look forward to seeing your solution!